Chapter 11 Review

Campbell Biology, AP Edition, 12e

Chapter 11 Review

A cheetah chases an impala.

Summary of Key Concepts

Advanced placement edition

Is all cell-to-cell communication regulated by the plasma membrane? What about apoptosis? **(BIG IDEA 3)**

To review key terms, go to the **Vocabulary Self-Quiz**.

Concept 11.1: External signals are converted to responses within the cell

- ********Signal transduction pathways** are crucial for many processes. Bacterial cells can sense the local density of bacterial cells (quorum sensing). Signaling during yeast cell mating has much in common with processes in multicellular organisms, suggesting an early evolutionary origin of signaling mechanisms.
- Local signaling by animal cells involves direct contact or the secretion of local regulators. For long-distance signaling, animal and plant cells use ****hormones**; animals also pass signals electrically.
- Like epinephrine, other hormones that bind to membrane receptors trigger a three-stage cell-signaling pathway:

An overview of cell signaling.

Binding of a signaling molecule from the extracellular fluid to a receptor in the plasma membrane triggers reception, transduction, and response within the cell cytoplasm.

1. Signal reception. The signaling molecule in the extracellular fluid binds to a receptor in the plasma membrane.

2. Signal transduction. In this diagram, three relay molecules show a signal transduction pathway where each relay molecule brings about a change in the next molecule in the pathway.

3. Cellular response. The final molecule in the pathway in step 2 activates a cellular response.

What determines whether *a cell responds to a hormone such as epinephrine? What determines* how *a cell responds to such a hormone?*

Concept 11.2: Signal reception: A signaling molecule binds to a receptor, causing it to change shape

- The binding between signaling molecule (****__ligand__) and receptor is highly specific. A specific shape change in a receptor is often the initial transduction of the signal.
- Three of the major types of cell-surface transmembrane receptors are the following: (1) **G protein-coupled receptors (GPCRs)** work with cytoplasmic **G proteins**. Ligand binding activates the receptor, which then activates a specific G protein, which activates yet another protein, thus propagating the signal. (2) **Receptor tyrosine kinases (RTKs)** react to the binding of signaling molecules by forming dimers and then adding phosphate groups to tyrosines on the cytoplasmic part of the other monomer making up the dimer. Relay proteins in the cell can then be activated by binding to different phosphorylated tyrosines, allowing this receptor to trigger several pathways at once. (3) **Ligand-gated ion channels** open or close in response to binding by specific signaling molecules, regulating the flow of specific ions across the membrane.
- The activity of all three types of receptors is crucial; abnormal GPCRs and RTKs are associated with many human diseases.
- Intracellular receptors are cytoplasmic or nuclear proteins. Signaling molecules that are hydrophobic or small enough to cross the plasma membrane bind to these receptors inside the cell.

How are the structures of a GPCR and an RTK similar? How does initiation of signal transduction differ for these two types of receptors?

Concept 11.3: Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell

- At each step in a signal transduction pathway, the signal is transduced into a different form, which commonly involves a shape change in a protein. Many signal transduction pathways include ****__phosphorylation cascades__, in which a series of ****__protein kinases__ each add a phosphate group to the next one in line, activating it. Enzymes called ****__protein phosphatases__ remove the phosphate groups. The balance between phosphorylation and dephosphorylation regulates the activity of proteins involved in the sequential steps of a signal transduction pathway.
- ****__Second messengers__, such as the small molecule ****__cyclic AMP (cAMP)__ and the ion Ca2+, diffuse readily through the cytosol and thus help broadcast signals quickly. Many G proteins activate ****__adenylyl cyclase__, which makes cAMP from ATP. Cells use Ca2+ as a second messenger in GPCR and RTK pathways. The tyrosine kinase pathways can also involve two other

second messengers, ****__diacylglycerol (DAG)__ and ****__inositol trisphosphate__IP3. IP3 can trigger a subsequent increase in Ca2+ level.

What is the difference between a protein kinase and a second messenger? Can both operate in the same signal transduction pathway?

Concept 11.4: Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities

- Some pathways lead to a nuclear response: Specific genes are turned on or off by activated transcription factors. In others, the response involves cytoplasmic regulation.
- Cellular responses are not simply on or off; they are regulated at many steps. Each protein in a signaling pathway amplifies the signal by activating multiple copies of the next component; for long pathways, the total amplification may be over a millionfold. The combination of proteins in a cell confers specificity in the signals it detects and the responses it carries out. ****__Scaffolding proteins__ increase signaling efficiency. Pathway branching further helps the cell coordinate signals and responses. Signal response can be terminated quickly because ligand binding is reversible.

What mechanisms in the cell terminate its response to a signal and maintain its ability to respond to new signals?

Concept 11.5: Apoptosis requires integration of multiple cell-signaling pathways

- ****__Apoptosis__is a type of programmed cell death in which cell components are disposed of in an orderly fashion. Studies of the soil worm *Caenorhabditis elegans* clarified molecular details of the relevant signaling pathways. A death signal leads to activation of caspases and nucleases, the main enzymes involved in apoptosis.
- Several apoptotic signaling pathways with related proteins exist in the cells of humans and other mammals, triggered in different ways. Signals eliciting apoptosis can originate from outside or inside the cell.

What is an explanation for the similarities between genes in yeasts, nematodes, and mammals that control apoptosis?

*For suggested answers, see*Appendix A.**

Test Your Understanding

For more multiple-choice questions, go to the **Practice Test**.

Levels 1-2: Remembering/Understanding

- Binding of a signaling molecule to which type of receptor leads directly to a change in the distribution of substances on opposite sides of the membrane?
- intracellular receptor
- G protein-coupled receptor
- phosphorylated receptor tyrosine kinase dimer
- ligand-gated ion channel
- The activation of receptor tyrosine kinases is characterized by
- dimerization and phosphorylation.
- dimerization and IP3 binding.
- a phosphorylation cascade.
- GTP hydrolysis.
- Lipid-soluble signaling molecules, such as aldosterone, cross the membranes of all cells but affect only target cells because
- only target cells retain the appropriate DNA segments.
- intracellular receptors are present only in target cells.
- only target cells have enzymes that break down aldosterone.
- only in target cells is aldosterone able to initiate the phosphorylation cascade that turns genes on.
- Consider this pathway: epinephrine → G protein-coupled receptor → G protein → adenylyl cyclase → cAMP. Identify the second messenger.
- cAMP
- G protein
- GTP
- adenylyl cyclase
- Which of the following occurs during apoptosis?

- lysis of the cell
- direct contact between signaling cells
- fragmentation of the DNA
- release of proteases outside the cell

Levels 3-4: Applying/Analyzing

- Which observation suggested to Sutherland the involvement of a second messenger in epinephrine's effect on liver cells?
- Enzymatic activity was proportional to the amount of calcium added to a cell-free extract.
- Receptor studies indicated that epinephrine was a ligand.
- Glycogen breakdown was observed only when epinephrine was administered to intact cells.
- Glycogen breakdown was observed only when epinephrine and glycogen phosphorylase were mixed.
- Protein phosphorylation is commonly involved with which of the following?
- ligand binding by receptor tyrosine kinases.
- activation of G protein-coupled receptors.
- activation of protein kinase molecules.
- release of Ca2+ from the ER lumen.

Levels 5-6: Evaluating/Creating

AP Assessment Questions

- **DRAW IT***Science Practice 2* **Draw** the following apoptotic pathway, which operates in human immune cells. A death signal is received when a molecule called Fas binds its cell-surface receptor. The binding of many Fas molecules to receptors causes receptor clustering. The intracellular regions of the receptors, when together, bind proteins called adaptor proteins. These in turn bind to inactive molecules of caspase-8, which become activated and then activate caspase-3. Once activated, caspase-3 initiates apoptosis.
- **CONNECT TO BIG IDEA 1 Identify** the evolutionary mechanisms that might account for the origin and persistence of cell-to-cell signaling systems in prokaryotes.

- **SCIENTIFIC INQUIRY*Science Practice 6*** Epinephrine initiates a signal transduction pathway that produces cyclic AMP (cAMP) and leads to the breakdown of glycogen to glucose, a major energy source for cells. But glycogen breakdown is only part of the fight-or-flight response that epinephrine brings about; the overall effect on the body includes an increase in heart rate and alertness, as well as a burst of energy. Given that caffeine blocks the activity of cAMP phosphodiesterase, **propose** a mechanism by which caffeine ingestion leads to heightened alertness and sleeplessness.
- **SCIENCE, TECHNOLOGY, AND SOCIETY** The aging process is thought to be initiated at the cellular level. Among the changes that can occur after a certain number of cell divisions is the loss of a cell's ability to respond to growth factors and other signals. Much research into aging is aimed at understanding such losses, with the ultimate goal of extending the human life span. Not everyone, however, agrees that this is a desirable goal. If life expectancy were greatly increased, **discuss** what might be the social and ecological consequences.
- **CONNECT TO BIG IDEA 4** The properties of life emerge at the biological level of the cell. The highly regulated process of apoptosis is not simply the destruction of a cell; it is also an emergent property. **Write** a short essay (about 100–150 words) that briefly explains the role of apoptosis in the development and proper functioning of an animal, and **describe** how this form of programmed cell death is a process that emerges from the orderly integration of signaling pathways.
- **SYNTHESIZE YOUR KNOWLEDGE*Science Practice 6***

A bowl of regular potato chips and a bowl of taco-flavored chips.

- There are five basic tastes—sour, salty, sweet, bitter, and "umami." Salt is detected when the concentration of salt outside of a taste bud cell is higher than that inside of it, and ion channels allow the passive leakage of Na+ into the cell. The resulting change in membrane potential (see Concept 7.4) sends the "salty" signal to the brain. Umami is a savory taste generated by glutamate (glutamic acid, found in monosodium glutamate, or MSG), which is used as a flavor enhancer in foods such as taco-flavored tortilla chips. The glutamate receptor is a GPCR, which, when bound, initiates a signaling pathway that ends with a cellular response, perceived by you as "taste." If you eat a regular potato chip and then rinse your mouth, you will no longer taste salt. But if you eat a flavored tortilla chip and then rinse, the taste persists. (Try it!) **Propose** a possible explanation for this difference.

*For selected answers, see*Appendix A.**

Cells in various stages of division, with chromosomes and cytoskeleton elements visible.

Chapter12

The Cell Cycle

Figure 12.1

A multicellular organism starts out as a single cell that divides into two. Those two cells then divide into four, as shown in these fluorescence micrographs of a marine worm embryo. Cell division continues throughout an organism's life, for growth or to replace worn-out or damaged cells. Each time a cell divides in this way, it is crucial that the daughter cells be genetically identical to the parent cell.

A chart asks, how does one parent cell give rise to two genetically identical daughter cells?

Begin with a parent cell. Interphase: The cell grows; in preparation for cell division, the chromosomes are duplicated, with the genetic material (D N A) copied precisely.

Mitosis: The chromosome copies are separated from each other and moved to opposite ends of the cell.

Cytokinesis: The cell divides into two daughter cells, genetically identical to each other and to the parent cell. The daughter cells may go on to divide, repeating the cycle.

Key Concepts

- **12.1Most cell division results in genetically identical daughter cells****
- **12.2The mitotic phase alternates with interphase in the cell cycle****
- **12.3The eukaryotic cell cycle is regulated by a molecular control system****

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The continuity of life depends on a cell cycle in which genetic information from a parent cell is passed on to daughter cells. In unicellular organisms, mitosis results in transmission of a complete genome through asexual reproduction **(BIG IDEA 3)**. For multicellular organisms, mitosis ensures the growth of an individual, given adequate energy **(BIG IDEA 2)**.

Study Tip

Make a visual study guide: Figure 12.1 presents the events of the cell cycle as a simplified linear diagram. As you learn more about the cell cycle, draw a detailed linear diagram of the stages of interphase, mitosis,

and cytokinesis. Include explanatory labels. Add the circular chart from Figure 12.6 and show how the two diagrams are related.

The first part of the cell cycle, interphase, is hand drawn. During interphase, G 1 phase, metabolic activity and growth occurs.

Get Ready for This Chapter

In this chapter, you will learn about the process by which cells divide and the implications of cell division as related to cancer. Before beginning this chapter, you should be able to:

- Describe the structure of the nucleus and the chromosomes in it (see Concept 6.3 and Figure 6.9).
- Describe the structure and function of microtubules and microfilaments (see Concept 6.6, Table 6.1, Figure 6.21, and Figure 6.26).
- Explain cell signaling, including reception of the signal, signal transduction, and the cellular response (see Concept 11.1).
- Describe how phosphorylation affects proteins (see Concept 11.3).
- The following questions provide a quick check of your basic knowledge in these areas.

1. What is the best description of the structure of a chromosome in the nucleus of a non-dividing cell?

```
Two long DNA molecules wound around each other, associated with many proteins.
```

That's incorrect. Try again.

Incorrect. Review the following material: Within the nucleus of a non-dividing cell, the DNA is organized into discrete units called chromosomes, structures that carry the genetic information. Each chromosome contains one long DNA molecule associated with many proteins.

```
Two long DNA molecules wound around each other, with no proteins.
```

That's incorrect. Try again.

Incorrect. Review the following material: Within the nucleus of a non-dividing cell, the DNA is organized into discrete units called chromosomes, structures that carry the genetic information. Each chromosome contains one long DNA molecule associated with many proteins.

```
One long DNA molecule, associated with many proteins.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Within the nucleus of a non-dividing cell, the DNA is organized into discrete units called chromosomes, structures that carry the genetic information. Each chromosome contains one long DNA molecule associated with many proteins.

```
One long DNA molecule, with no proteins.
```

That's incorrect. Try again.

Incorrect. Review the following material: Within the nucleus of a non-dividing cell, the DNA is organized into discrete units called chromosomes, structures that carry the genetic information. Each chromosome contains one long DNA molecule associated with many proteins.

2. Which of the following statements best describes microtubules?

```
Microtubules are fibrous proteins coiled into cables
that maintain cell shape and anchor the nucleus.
```

That's incorrect. Try again.

Incorrect. Review the following material: Microtubules—hollow rods constructed from globular proteins called tubulins—shape and support the cell and serve as tracks along which organelles equipped with motor proteins can move. Microfilaments are made up of subunits of actin, and function in muscle contraction and changes in cell shape. Intermediate filaments are fibrous proteins coiled into cables that maintain cell shape and anchor the nucleus.

```
Microtubules are permanent cellular structures that help
the cell to move from place to place.
```

That's incorrect. Try again.

Incorrect. Review the following material: Microtubules—hollow rods constructed from globular proteins called tubulins—shape and support the cell and serve as tracks along which organelles equipped with motor proteins can move. Microfilaments are made up of subunits of actin, and function in muscle contraction and changes in cell shape. Intermediate filaments are fibrous proteins coiled into cables that maintain cell shape and anchor the nucleus.

```
Microtubules are made up of subunits of actin, and
function in muscle contraction and changes in cell shape.
```

That's incorrect. Try again.

Incorrect. Review the following material: Microtubules—hollow rods constructed from globular proteins called tubulins—shape and support the cell and serve as tracks along which organelles equipped with motor proteins can move. Microfilaments are made up of subunits of actin, and function in muscle contraction and changes in cell shape. Intermediate filaments are fibrous proteins coiled into cables that maintain cell shape and anchor the nucleus.

```
Microtubules are made up of subunits of tubulin, and are
structures along which substances are transported in the cell.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Microtubules—hollow rods constructed from globular proteins called tubulins—shape and support the cell and serve as tracks along which organelles equipped with motor proteins can move. Microfilaments are made up of subunits of actin, and function in muscle contraction and changes in cell shape. Intermediate filaments are fibrous proteins coiled into cables that maintain cell shape and anchor the nucleus.

3. How can a protein outside the cell cause events to happen inside the cell?

```
A protein outside the cell can bind to a receptor
protein on the cell surface, causing it to change shape and
sending a signal inside the cell.
```

That's correct.

Correct! If you had trouble with this question, review the following material: The hydrophobic interior of the membrane impedes direct passage of proteins into the cell. Reception, the target cell’s detection of a

signaling molecule coming from outside the cell, occurs when a protein binds to a receptor protein located at the cell's surface, changing the shape of the receptor protein and initiating transduction. Transduction occurs within the cell and typically involves several steps, converting the external signal into a form that brings about a specific cellular response inside the cell.

```
A protein can pass through the membrane and into the cell, causing signaling inside the cell.
```

That's incorrect. Try again.

Incorrect. Review the following material: The hydrophobic interior of the membrane impedes direct passage of proteins into the cell. Reception, the target cell's detection of a signaling molecule coming from outside the cell, occurs when a protein binds to a receptor protein located at the cell's surface, changing the shape of the receptor protein and initiating transduction. Transduction occurs within the cell and typically involves several steps, converting the external signal into a form that brings about a specific cellular response inside the cell.

```
A protein outside the cell can cause the hydrogen ion concentration outside the cell to be higher, which allows substances to pass into the cell and transduce a signal.
```

That's incorrect. Try again.

Incorrect. Review the following material: The hydrophobic interior of the membrane impedes direct passage of proteins into the cell. Reception, the target cell's detection of a signaling molecule coming from outside the cell, occurs when a protein binds to a receptor protein located at the cell's surface, changing the shape of the receptor protein and initiating transduction. Transduction occurs within the cell and typically involves several steps, converting the external signal into a form that brings about a specific cellular response inside the cell.

```
A protein outside the cell can phosphorylate a protein inside the cell, which transduces a signal and causes a response.
```

That's incorrect. Try again.

Incorrect. Review the following material: The hydrophobic interior of the membrane impedes direct passage of proteins into the cell. Reception, the target cell's detection of a signaling molecule coming from outside the cell, occurs when a protein binds to a receptor protein located at the cell's surface, changing the shape of the receptor protein and initiating transduction. Transduction occurs within the cell and typically involves several steps, converting the external signal into a form that brings about a specific cellular response inside the cell.

4. What effect does phosphorylating a protein have on that protein?

```
        Phosphorylation changes the shape of the protein, most
often activating it.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Protein phosphorylation causes a shape change in the phosphorylated protein. The shape change results from the interaction of the newly added phosphate groups with charged or polar amino acids on the protein being phosphorylated. The shape change alters the function of the protein, most often activating it, but in some cases phosphorylation instead decreases the activity of the protein.

```
        Phosphorylation may have no effect at all on the
protein.
```

That's incorrect. Try again.

Incorrect. Review the following material: Protein phosphorylation causes a shape change in the phosphorylated protein. The shape change results from the interaction of the newly added phosphate groups with charged or polar amino acids on the protein being phosphorylated. The shape change alters the function of the protein, most often activating it, but in some cases phosphorylation instead decreases the activity of the protein.

```
        Phosphorylation adds one or more phosphorus atoms to the
protein, allowing it to perform a particular function.
```

That's incorrect. Try again.

Incorrect. Review the following material: Protein phosphorylation causes a shape change in the phosphorylated protein. The shape change results from the interaction of the newly added phosphate groups

with charged or polar amino acids on the protein being phosphorylated. The shape change alters the function of the protein, most often activating it, but in some cases phosphorylation instead decreases the activity of the protein.

Phosphorylation adds energy to the protein, allowing it to form ATP from ADP and Pi.

That's incorrect. Try again.

Incorrect. Review the following material: Protein phosphorylation causes a shape change in the phosphorylated protein. The shape change results from the interaction of the newly added phosphate groups with charged or polar amino acids on the protein being phosphorylated. The shape change alters the function of the protein, most often activating it, but in some cases phosphorylation instead decreases the activity of the protein.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 12.1 Explain how mitosis results in genetically identical cells, using the terms chromosome, chromatid, and chromatin.
- 12.2 Describe the events of interphase and the M phase.
- 12.3 Identify three checkpoints that control the cell cycle and explain how they operate.

Concept 12.1: Most cell division results in genetically identical daughter cells

The ability of organisms to produce more of their own kind is the one characteristic that best distinguishes living things from nonliving matter. This unique capacity to procreate, like all biological functions, has a cellular basis. The continuity of life is based on the reproduction of cells, or **cell division**.

Key Roles of Cell Division

Cell division plays several important roles in life. When a prokaryotic cell divides, it is actually reproducing because the process gives rise to a new organism (another cell). The same is true of any unicellular eukaryote, such as the amoeba shown in **Figure 12.2a**. As for multicellular eukaryotes, cell division enables each of these organisms to develop from a single cell—the fertilized egg. A two-celled embryo, the first stage in this process, is shown in **Figure 12.2b**. And cell division continues to function in renewal and repair in

fully grown multicellular eukaryotes, replacing cells that die from accidents or normal wear and tear. For example, dividing cells in your bone marrow continuously make new blood cells **(Figure 12.2c)**.

Figure 12.2

The functions of cell division.

The functions of cell division: reproduction, growth and development, and tissue renewal.

Part a. Asexual reproduction. An amoeba, a single-celled eukaryote, is divided into two cells. Each new cell will be an individual organism.

Part b. Growth and development. This micrograph shows a sand dollar embryo shortly after the fertilized egg is divided, forming two cells.

Part c. Tissue renewal. These dividing bone marrow cells will give rise to new blood cells.

Video

Cell Division in a Sea Urchin Embryo

image

Play

image

Play

The reproduction of a cell, with all of its complexity, cannot occur by a mere pinching in half; a cell is not like a soap bubble that simply enlarges and splits in two. In both prokaryotes and eukaryotes, a crucial function of most cell divisions is the distribution of identical genetic material—DNA—to two daughter cells. (The exception is meiosis, the special type of eukaryotic cell division that can produce sperm and eggs.) What is most remarkable about cell division is the accuracy with which the DNA is passed from one generation of cells to the next. A dividing cell replicates its DNA, distributes the two copies to opposite ends of the cell, and then splits into daughter cells.

Cellular Organization of the Genetic Material

A cell's DNA, its genetic information, is called its ****genome**. Although a prokaryotic genome is often a single DNA molecule, eukaryotic genomes usually consist of a number of DNA molecules. The overall length of DNA in a eukaryotic cell is enormous. A typical human cell, for example, has about 2 m of DNA—a length about 250,000 times greater than the cell's diameter. Before the cell can divide to form genetically identical

daughter cells, all of this DNA must be copied, or replicated, and then the two copies must be separated so that each daughter cell ends up with a complete genome.

The replication and distribution of so much DNA are manageable because the DNA molecules are packaged into structures called ****chromosomes** (from the Greek *chroma*, color, and *soma*, body), so named because they take up certain dyes used in microscopy **(Figure 12.3)**. Each eukaryotic chromosome consists of one very long, linear DNA molecule associated with many proteins (see Figure 6.9). The DNA molecule carries several hundred to a few thousand genes, the units of information that specify an organism's inherited traits. The associated proteins maintain the structure of the chromosome and help control the activity of the genes. Together, the entire complex of DNA and proteins that is the building material of chromosomes is referred to as ****chromatin**. As you will soon see, the chromatin of a chromosome varies in its degree of condensation during the process of cell division.

Figure 12.3

Eukaryotic chromosomes.

Chromosomes visible as thick dark lines. Scale: 20 micrometers.

Chromosomes (stained purple) are visible within the nucleus of this cell from an African blood lily. The thinner red threads in the surrounding cytoplasm are the cytoskeleton. The cell is preparing to divide (LM).

Every eukaryotic species has a characteristic number of chromosomes in each cell's nucleus. For example, the nuclei of human **somatic cells** (all body cells except the reproductive cells) each contain 46 chromosomes, made up of two sets of 23, one set inherited from each parent. Reproductive cells, or **gametes**—such as sperm and eggs—have half as many chromosomes as somatic cells; in our example, human gametes have one set of 23 chromosomes. The number of chromosomes in somatic cells varies widely among species: 18 in cabbage plants, 48 in chimpanzees, 56 in elephants, 90 in hedgehogs, and 148 in one species of alga. We'll now consider how these chromosomes behave during cell division.

Distribution of Chromosomes During Eukaryotic Cell Division

When a cell is not dividing, and even as it replicates its DNA in preparation for cell division, each chromosome is in the form of a long, thin chromatin fiber. After DNA replication, however, the chromosomes condense as a part of cell division: Each chromatin fiber becomes densely coiled and folded, making the chromosomes much shorter and so thick that we can see them with a light microscope.

Each duplicated chromosome consists of two **sister chromatids**, which are joined copies of the original chromosome **(Figure 12.4)**. The two chromatids, each containing an identical DNA molecule, are often attached all along their lengths by protein complexes called *cohesins*; this attachment is known as *sister*

chromatid cohesion. Each sister chromatid has a **centromere**, a region made up of repetitive sequences in the chromosomal DNA where the chromatid is attached most closely to its sister chromatid. This attachment is mediated by proteins that recognize and bind to the centromeric DNA; other bound proteins condense the DNA, giving the duplicated chromosome a narrow "waist." The portion of a chromatid to either side of the centromere is referred to as an *arm* of the chromatid. (An unduplicated chromosome has a single centromere, distinguished by the proteins that bind there, and two arms.)

Figure 12.4

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A highly condensed, duplicated human chromosome. (SEM)
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Image of a duplicated human chromosome in the form of sister chromatids (two thick strands joined near the center in a pinched-in waist region called the centromere). Scale: 0.5 micrometers.

DRAW IT *Circle one sister chromatid of this chromosome.*

*For suggested answer, see**Appendix A.*

Later in the cell division process, the two sister chromatids of each duplicated chromosome separate and move into two new nuclei, one forming at each end of the cell. Once the sister chromatids separate, they are no longer called sister chromatids but are considered individual chromosomes; this is the step that essentially doubles the number of chromosomes during cell division. Thus, each new nucleus receives a collection of chromosomes identical to that of the parent cell **(Figure 12.5)**. **Mitosis**, the division of the genetic material in the nucleus, is usually followed immediately by **cytokinesis**, the division of the cytoplasm. One cell has become two, each the genetic equivalent of the parent cell.

Figure 12.5

```
Chromosome duplication and distribution during cell
division.
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One of the multiple chromosomes in a eukaryotic cell shown here in condensed form, not yet duplicated. Normally would be a long, thin chromatin fiber containing one D N A molecule and proteins.

Step 1. A single chromosome in a cell is shown as a thick line with a pinched-in waist region called the centromere. The chromosome arm is the region on either side of the centromere. A double helix representing D N A is drawn next to the chromosome, illustrating that the chromosome consists of packaged

chromosomal D N A molecules. Arrows pointing to Figure 12.b indicate chromosome duplication (including D N A replication) and condensation.

Step 2. Once duplicated, a chromosome consists of two sister chromatids connected along their entire lengths by sister chromatid cohesion. Each chromatid contains a copy of the D N A molecule. Cell now contains two thick lines representing sister chromatids joined at the centromere. Two double helices representing D N A molecules are also shown, indicating that the D N A has been duplicated in forming sister chromatids. Arrows point to Figure 12.5 c, indicating the separation of sister chromatids into two chromosomes.

Step 3. Molecular and mechanical processes separate the sister chromatids into two chromosomes and distribute them to two daughter cells. Two cells are drawn, each with a single chromosome (thick line). Each chromosome contains identical D N A.

How many chromatid arms does the chromosome in **STEP 2** *have? Identify the point in the figure where one chromosome becomes two.*

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Chromosome Duplication

image

Play

image

Play

From a fertilized egg, mitosis and cytokinesis produced the 37 trillion somatic cells that nowvmake up your body, and the same processes continue to generate new cells to replace dead and damaged ones. In contrast, you produce gametes—eggs or sperm—by a variation of cell division called *meiosis*, which yields daughter cells with only one set of chromosomes, half as many chromosomes as the parent cell. Meiosis in humans occurs only in special cells in the ovaries or testes (the gonads). Generating gametes, meiosis reduces the chromosome number from 46 (two sets) to 23 (one set). Fertilization fuses two gametes together and returns the chromosome number to 46 (two sets). Mitosis then conserves that number in every somatic cell nucleus of the new human individual. In Chapter 13, we will examine the role of meiosis in reproduction and inheritance in more detail. In the remainder of this chapter, we focus on mitosis and the rest of the cell cycle in eukaryotes.

Concept Check 12.1

- How many chromosomes are drawn in each part of Figure 12.5? (Ignore the micrograph in step 2.)
- **WHAT IF?** A chicken has 78 chromosomes in its somatic cells. How many chromosomes did the chicken inherit from each parent? How many chromosomes are in each of the chicken's gametes? How many chromosomes will be in each somatic cell of the chicken's offspring?

*For suggested answers, see*Appendix A.**

Concept 12.2: The mitotic phase alternates with interphase in the cell cycle

In 1882, a German anatomist named Walther Flemming developed dyes that allowed him to observe, for the first time, the behavior of chromosomes during mitosis and cytokinesis. (In fact, Flemming coined the terms *mitosis* and *chromatin*.) During the period between one cell division and the next, it appeared to Flemming that the cell was simply growing larger. But we now know that many critical events occur during this stage in the life of a cell.

CheckPoint

Counting Chromosomes

Check your understanding before you move on, with the CheckPoint: Counting Chromosomes.

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<p>iframes are not supported by your browser.</p>
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Phases of the Cell Cycle

Mitosis is just one part of the **cell cycle**, the life of a cell from the time it is first formed during division of a parent cell until its own division into two daughter cells **(Figure 12.6)**. (Biologists use the words *daughter* or *sister* in relation to cells, but this is not meant to imply gender.) In fact, the **mitotic phase (M)**, which includes both mitosis and cytokinesis, is usually the shortest part of the cell cycle. The mitotic phase alternates with a much longer stage called **interphase**, which often accounts for about 90% of the cycle. Interphase can be divided into three phases: the **G1 phase** ("first gap"), the **S phase** ("synthesis"), and the **G2 phase** ("second gap"). The G phases were misnamed as "gaps" when they were first observed because the cells appeared inactive, but we now know that intense metabolic activity and growth occur throughout

interphase. During all three phases of interphase, actually, a cell grows by producing proteins and cytoplasmic organelles such as mitochondria and endoplasmic reticulum. Duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during the S phase. (We'll explore synthesis, or replication, of DNA in Concept 16.2.) Thus, a cell grows G1, continues to grow as it copies its chromosomes (S), grows more as it completes preparations for cell division G2, and divides (M). The daughter cells may then repeat the cycle.

Figure 12.6

```
The cell cycle.
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Circular chart depicting the cell cycle.

Interphase takes up most of the circle chart and is divided into G 1, S (D N A synthesis), and G 2 phases, with G 1 being the largest slice. The mitotic phase (M) takes up a small slice of the chart and is divided into Mitosis and Cytokinesis. During mitosis, the distribution of chromosomes into two daughter nuclei occurs. During cytokinesis, the division of the cytoplasm, producing two daughter cells occurs. Each daughter cell can start a new cell cycle.

In a dividing cell, the mitotic (M) phase alternates with interphase, a growth period.

Animation

The Cell Cycle

image

image

**

A particular human cell might undergo one division in 24 hours. Of this time, the M phase would occupy less than 1 hour, while the S phase might occupy 10–12hours, or about half the cycle. The rest of the time would be apportioned between the G1 and G2 phases. The G2 phase usually takes 4–6hours; in our example, G1 would occupy about 5–6hours. G1 is the most variable in length in different types of cells. Some cells in a multicellular organism divide very infrequently or not at all. These cells spend their time in G1 (or a related phase called G0, to be discussed later in the chapter) doing their job in the organism—a cell of the pancreas secretes digestive enzymes, for example.

Mitosis is conventionally broken down into five stages: **prophase**, **prometaphase**, **metaphase**, **anaphase**, and **telophase**. Overlapping with the latter stages of mitosis, cytokinesis completes the mitotic

phase. **Figure 12.7** describes these stages in an animal cell. Study this figure thoroughly before proceeding to the next two sections, which examine mitosis and cytokinesis more closely.

Figure 12.7

Exploring Mitosis in an Animal Cell

Three stages of mitosis in an animal cell: G 2 of interphase, prophase, prometaphase. The stages of mitosis continued.

The steps in each phase are as follows. G 2 of interphase.

- A nuclear envelope encloses the nucleus.
- The nucleus contains one or more nucleoli (singular, nucleolus).
- Two centrosomes have formed by duplication of a single centrosome.
- Centrosomes are regions in animal cells that organize the microtubules of the spindle. Each centrosome contains two centrioles.
- Chromosomes, duplicated during the S phase, cannot be seen individually because they have not yet been condensed.

Prophase.

- The chromatin fibers become more tightly coiled, condensing into discrete chromosomes observable with a light microscope.
- The nucleoli disappear.
- Each duplicated chromosome appears as two identical sister chromatids joined at their centromeres and, in some species, all along their arms by cohesins (sister chromatid cohesion).
- The mitotic spindle (named for its shape) begins to form. It is composed of the centrosomes and the microtubules that extend from them. The radial arrays of shorter microtubules that extend from the centrosomes are called asters (stars).
- The centrosomes move away from each other, propelled partly by the lengthening microtubules between them.

Prometaphase.

- The nuclear envelope fragments.
- The microtubules extending from each centrosome can now invade the nuclear area.
- The chromosomes have become even more condensed.
- A kinetochore, a specialized protein structure, has now formed at the centromere of each chromatid (thus, two per chromosome).
- Some of the microtubules attach to the kinetochores, becoming kinetochore microtubules, which jerk the chromosomes back and forth.

- Non-kinetochore microtubules interact with those from the opposite pole of the spindle, lengthening the cell.

The steps in each phase continue as follows.

Metaphase.

- The centrosomes are now at opposite poles of the cell.
- The chromosomes have all arrived at the metaphase plate, a plane that is equidistant between the spindle's two poles. The chromosomes' centromeres lie at the metaphase plate.
- For each chromosome, the kinetochores of the sister chromatids are attached to kinetochore microtubules coming from opposite poles.

Anaphase.

- Anaphase is the shortest stage of mitosis, often lasting only a few minutes.
- Anaphase begins when the cohesin proteins are cleaved. This allows the two sister chromatids of each pair to part suddenly. Each chromatid thus becomes an independent chromosome.
- The two new daughter chromosomes begin moving toward opposite ends of the cell as their kinetochore microtubules shorten. Because these microtubules are attached at the centromere region, the centromeres are pulled ahead of the arms, moving at a rate of about 1 micrometer per minute.
- The cell elongates as the non-kinetochore microtubules lengthen.
- By the end of anaphase, the two ends of the cell have equivalent, and complete, collections of chromosomes.

Telophase.

- Two daughter nuclei form in the cell. Nuclear envelopes arise from the fragments of the parent cell's nuclear envelope and other portions of the endomembrane system.
- Nucleoli reappear.
- The chromosomes become less condensed.
- Any remaining spindle microtubules are depolymerized.
- Mitosis, the division of one nucleus into two genetically identical nuclei, is now complete.

Cytokinesis.

- The division of the cytoplasm is usually well under way by late telophase, so the two daughter cells appear shortly after the end of mitosis.
- In animal cells, cytokinesis involves the formation of a cleavage furrow, which pinches the cell in two.

How many molecules of DNA are in the prometaphase drawing? How many molecules per chromosome? How many double helices are there per chromosome? Per chromatid?

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Mitosis

image

image

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Video

Animal Mitosis (time-lapse)

image

image

The Mitotic Spindle: *A Closer Look*

Many of the events of mitosis depend on the **mitotic spindle**, which begins to form in the cytoplasm during prophase. This structure consists of fibers made of microtubules and associated proteins. While the mitotic spindle assembles, the other microtubules of the cytoskeleton partially disassemble, providing the material used to construct the spindle. The spindle microtubules elongate (polymerize) by incorporating more subunits of the protein tubulin (see Table 6.1) and shorten (depolymerize) by losing subunits.

In animal cells, the assembly of spindle microtubules starts at the **centrosome**,**** a subcellular region containing material that functions throughout the cell cycle to organize the cell's microtubules. (It is also a type of *microtubule-organizing center*.) A pair of centrioles is located at the center of the centrosome, but they are not essential for cell division: If the centrioles are destroyed with a laser microbeam, a spindle nevertheless forms during mitosis. In fact, centrioles are not even present in plant cells, which do form mitotic spindles.

During interphase in animal cells, the single centrosome duplicates, forming two centrosomes, which remain near the nucleus. The two centrosomes move apart during prophase and prometaphase of mitosis as spindle microtubules grow out from them. By the end of prometaphase, the two centrosomes, one at each pole of the spindle, are at opposite ends of the cell. An *aster*, a radial array of short microtubules, extends from each centrosome. The spindle includes the centrosomes, the spindle microtubules, and the asters.

Each of the two sister chromatids of a duplicated chromosome has a **kinetochore**, a structure made up of proteins that have assembled on specific sections of DNA at each centromere. The chromosome's two kinetochores face in opposite directions. During prometaphase, some of the spindle microtubules attach to the kinetochores; these are called *kinetochore microtubules*. (The number of microtubules attached to a kinetochore varies among species, from one microtubule in yeast cells to 40 or so in some mammalian cells.) When one of a chromosome's kinetochores is "captured" by microtubules, the chromosome begins to move toward the pole from which those microtubules extend. However, this movement comes to a halt as soon as microtubules from the opposite pole attach to the kinetochore on the other chromatid. What happens next is like a tug-of-war that ends in a draw. The chromosome moves first in one direction and then in the other, back and forth, finally settling midway between the two ends of the cell. At metaphase, the centromeres of all the duplicated chromosomes are on a plane midway between the spindle's two poles. This plane is called the **metaphase plate**, which is an imaginary plate rather than an actual cellular structure **(Figure 12.8)**. Meanwhile, microtubules that do not attach to kinetochores have been elongating, and by metaphase they overlap and interact with other *nonkinetochore microtubules* from the opposite pole of the spindle. By metaphase, the microtubules of the asters have also grown and are in contact with the plasma membrane. The spindle is now complete.

Figure 12.8

The mitotic spindle at metaphase.

Diagram and TEMs of cell in metaphase.

The kinetochores of each chromosome's two sister chromatids face in opposite directions. Here, each kinetochore is attached to a cluster of kinetochore microtubules (see TEM) extending from the nearest centrosome. Nonkinetochore microtubules overlap at the metaphase plate. The fluorescent micrograph is a rat kangaroo cell at metaphase.

Diagram: Six chromosomes in sister chromatid form lined up at metaphase place (an imaginary structure at the middle of the cell). Kinetochores of sister chromatids face the poles of the cells and attach to a cluster of kinetochore microtubules (drawn as thick cable-like structures). Non-kinetochore microtubules overlap at the metaphase plate. Centrosomes are labeled at the poles of the cells, where centrioles with asters are visible. Arrows from a kinetochore in one sister chromatid zoom in on a T E M of the kinetochore microtubules (scale: 0.5 micrometers). Another T E M (scale: 1 micrometer) shows thick chromosomes lined up in metaphase with the microtubules of the mitotic spindle visible.

DRAW IT *On the lower micrograph, draw a line indicating the position of the metaphase plate. Draw arrows showing the directions of chromosome movement when anaphase begins.*

*For suggested answer, see*Appendix A.**

Video

Spindle Formation During Mitosis

image

image

Animation

Mitosis

image

image

The structure of the spindle correlates well with its function during anaphase. Anaphase begins suddenly when the cohesins holding together the sister chromatids of each chromosome are cleaved by an enzyme called *separase*. Once separated, the chromatids become individual chromosomes that move toward opposite ends of the cell.

How do the kinetochore microtubules function in this poleward movement of chromosomes? Apparently, two mechanisms are in play, both involving motor proteins. (To review how motor proteins move an object along a microtubule, see Figure 6.21.) Results of a cleverly designed experiment suggested that motor proteins on the kinetochores "walk" the chromosomes along the microtubules, which depolymerize at their kinetochore ends after the motor proteins have passed **(Figure 12.9)**. (This is referred to as the "Pacman" mechanism because of its resemblance to the arcade game character that moves by eating all the dots in its path.) However, other researchers, working with different cell types or cells from other species, have shown that chromosomes are "reeled in" by motor proteins at the spindle poles and that the microtubules depolymerize after they pass by these motor proteins at the poles. The general consensus now is that both mechanisms are used and that their relative contributions vary among cell types.

Figure 12.9

Inquiry: At which end do kinetochore microtubules shorten during anaphase?

Experiment

Gary Borisy and colleagues at the University of Wisconsin wanted to determine whether kinetochore microtubules depolymerize at the kinetochore end or the pole end as chromosomes move toward the poles during mitosis. First they labeled the microtubules of a pig kidney cell in early anaphase with a yellow fluorescent dye. (Nonkinetochore microtubules are not shown.)

A diagram of a cell's kinetochore microtubules has kinetochore ends and spindle poles labeled and microtubules colored yellow.

Then they marked a region of the kinetochore microtubules between one spindle pole and the chromosomes by using a laser to eliminate the fluorescence from that region, while leaving the microtubules intact (see below). As anaphase proceeded, they monitored the changes in microtubule length on either side of the non-fluorescent mark.

A diagram of a cell's kinetochore microtubules has a darkened region marked on the microtubules between the spindle pole and kinetochore end.

Results

As the chromosomes moved poleward, the microtubule segments on the kinetochore side of the mark shortened, while those on the spindle pole side stayed the same length.

A diagram of a cell's kinetochore microtubules. The microtubules are shortened. The darkened areas are closer to the kinetochore ends than the poles.

Conclusion

During anaphase in this cell type, chromosome movement is correlated with kinetochore microtubules shortening at their kinetochore ends and not at their spindle pole ends. This experiment supports the hypothesis that during anaphase, a chromosome is walked along a microtubule as the microtubule depolymerizes at its kinetochore end, releasing tubulin subunits.

A diagram shows chromosome movement along a microtubule toward the pole end as the microtubule depolymerizes at its kinetochore end, releasing tubulin subunits.

Data from G. J. Gorbsky, P. J. Sammak, and G. G. Borisy, Chromosomes move poleward in anaphase along stationary microtubules that coordinately disassemble from their kinetochore ends,*Journal of Cell Biology* 104:9–18 (1987).

WHAT IF? *If this experiment had been done on a cell type in which "reeling in" at the poles was the main cause of chromosome movement, how would the mark have moved relative to the poles? How would the microtubule portions on either side of the mark have changed?*

*For suggested answer, see**Appendix A.*

Animation

Microtubule Depolymerization

image

image

In a dividing animal cell, the nonkinetochore microtubules are responsible for elongating the whole cell during anaphase. Nonkinetochore microtubules from opposite poles overlap each other extensively during metaphase (see Figure 12.8). During anaphase, the region of overlap is reduced as motor proteins attached to the microtubules walk them away from one another, using energy from ATP. As the microtubules push apart from each other, their spindle poles are pushed apart, elongating the cell. At the same time, the microtubules lengthen somewhat by the addition of tubulin subunits to their overlapping ends. As a result, the microtubules continue to overlap.

At the end of anaphase, duplicate groups of chromosomes have arrived at opposite ends of the elongated parent cell. Nuclei re-form during telophase. Cytokinesis generally begins during anaphase or telophase, and the spindle eventually disassembles by depolymerization of microtubules.

Cytokinesis: *A Closer Look*

In animal cells, cytokinesis occurs by a process known as **cleavage**. The first sign of cleavage is the appearance of a **cleavage furrow**, a shallow groove in the cell surface near the old metaphase plate **(Figure 12.10a)**. On the cytoplasmic side of the furrow is a contractile ring of actin microfilaments associated with molecules of the protein myosin. The actin microfilaments interact with the myosin molecules, causing the ring to contract. The contraction of the dividing cell's ring of microfilaments is like the pulling of a drawstring. The cleavage furrow deepens until the parent cell is pinched in two, producing two completely separated cells, each with its own nucleus and its own share of cytosol, organelles, and other subcellular structures.

Figure 12.10

Cytokinesis in animal and plant cells.

Cytokinesis in animal and plant cells.

Part a. S E M shows groove between two cell halves (scale: 100 micrometers). Below: diagram of cleavage; furrow develops as a result of a contractile ring of microfilaments, resulting in two separate daughter cells.

Part b. T E M shows vesicles forming cell plates between newly forming cells (scale: 1 micrometer); parent cell wall visible. Diagram: vesicles forming cell plate going to new cell wall between daughter cells.

Animation

Cytokinesis

image

image

Video

Cytokinesis in an Animal Cell

image

image

Cytokinesis in plant cells, which have cell walls, is markedly different. There is no cleavage furrow. Instead, during telophase, vesicles derived from the Golgi apparatus move along microtubules to the middle of the cell, where they coalesce, producing a **cell plate**. Cell wall materials carried in the vesicles collect inside the cell plate as it grows **(Figure 12.10b)**. The cell plate enlarges until its surrounding membrane fuses with the plasma membrane along the perimeter of the cell. Two daughter cells result, each with its own plasma membrane. Meanwhile, a new cell wall arising from the contents of the cell plate forms between the daughter cells.

Figure 12.11 is a series of micrographs of a dividing plant cell. Examining this figure will help you review mitosis and cytokinesis.

Figure 12.11

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    Mitosis in a plant cell.
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The five stages of mitosis in a plant cell: prophase, prometaphase, metaphase, anaphase, and telophase. Scale for micrographs: 10 micrometers.

These light micrographs show mitosis in cells of an onion root.

1. Prophase. The chromosomes are condensing and the nucleolus is beginning to disappear. Although not yet visible in the micrograph, the mitotic spindle is starting to form.

2. Prometaphase. Discrete chromosomes are now visible; each consists of two aligned, identical sister chromatids. Later in pro-metaphase, the nuclear envelope will fragment.
3. Metaphase. The spindle is complete, and the chromosomes, attached to microtubules at their kinetochores, are all at the metaphase plate.
4. Anaphase. The chromatids of each chromosome have separated, and the daughter chromosomes are moving to the ends of the cell as their kinetochore microtubules shorten.
5. Telophase. Daughter nuclei are forming. Meanwhile, cytokinesis has started: The cell plate, which will divide the cytoplasm in two, is growing toward the perimeter of the parent cell.

Binary Fission in Bacteria

Prokaryotes (bacteria and archaea) can undergo a type of reproduction in which the cell grows to roughly double its size and then divides to form two cells. The term **binary fission**, meaning "division in half," refers to this process and to the asexual reproduction of single-celled eukaryotes, such as the amoeba in Figure 12.2a. However, the process in eukaryotes involves mitosis, while that in prokaryotes does not.

In bacteria, most genes are carried on a single bacterial chromosome that consists of a circular DNA molecule and associated proteins. Although bacteria are smaller and simpler than eukaryotic cells, the challenge of replicating their genomes in an orderly fashion and distributing the copies equally to two daughter cells is still formidable. For example, when it is fully stretched out, the chromosome of the bacterium *Escherichia coli* is about 500 times as long as the cell. For such a long chromosome to fit within the cell, it must be highly coiled and folded.

In some bacteria, the process of cell division is initiated when the DNA of the bacterial chromosome begins to replicate at a specific place on the chromosome called the **origin of replication**, producing two origins. As the chromosome continues to replicate, one origin moves rapidly toward the opposite end of the cell **(Figure 12.12)**. While the chromosome is replicating, the cell elongates. When replication is complete and the bacterium has reached about twice its initial size, proteins cause its plasma membrane to pinch inward, dividing the parent bacterial cell into two daughter cells. In this way, each cell inherits a complete genome.

Figure 12.12

Bacterial cell division by binary fission.

Bacterial cell division by binary fission.

The bacterium E. coli, shown here, has a single, circular chromosome.

Step 1. Chromosome replication begins. Soon after, one copy of the origin moves rapidly toward the other end of the cell by a mechanism involving an actin-like protein.

Step 2. Replication continues. One copy of the origin is now at each end of the cell. Meanwhile, the cell elongates. Bacterial cell are longer than the original cell.

Step 3. Replication finishes. The plasma membrane is pinched inward by a tubulin-like protein, and a new cell wall is deposited.

Step 4. Two daughter cells result. Each cell has a copy of the bacterial chromosome with the origin marked.

Animation

Cell Division in Bacteria

Using the techniques of modern DNA technology to tag the origins of replication with molecules that glow green in fluorescence microscopy (see Figure 6.3), researchers have directly observed the movement of bacterial chromosomes. This movement is reminiscent of the poleward movements of the centromere regions of eukaryotic chromosomes during anaphase of mitosis, but bacteria don't have visible mitotic spindles or even microtubules. In most bacterial species studied, the two origins of replication end up at opposite ends of the cell or in some other very specific location, possibly anchored there by one or more proteins. How bacterial chromosomes move and how their specific location is established and maintained are active areas of research. Several proteins that play important roles have been identified. Polymerization of one protein resembling eukaryotic actin apparently functions in bacterial chromosome movement during cell division, and another protein that is related to tubulin helps pinch the plasma membrane inward, separating the two bacterial daughter cells.

The Evolution of Mitosis

EVOLUTION Given that prokaryotes preceded eukaryotes on Earth by more than a billion years, we might hypothesize that mitosis evolved from simpler prokaryotic mechanisms of cell reproduction. The fact that some of the proteins involved in bacterial binary fission are related to eukaryotic proteins that function in mitosis supports that hypothesis.

As eukaryotes with nuclear envelopes and larger genomes evolved, the ancestral process of binary fission, seen today in bacteria, somehow gave rise to mitosis. Variations on cell division exist in different groups of organisms. These variant processes may be similar to mechanisms used by ancestral species and thus may resemble steps in the evolution of mitosis from a binary fission-like process presumably carried out by very early bacteria. Possible intermediate stages are suggested by two unusual types of nuclear division found today in certain unicellular eukaryotes—dinoflagellates, diatoms, and some yeasts **(Figure 12.13)**. These two modes of nuclear division are thought to be cases where ancestral mechanisms have remained relatively

unchanged over evolutionary time. In both types, the nuclear envelope remains intact, in contrast to what happens in most eukaryotic cells. Keep in mind, however, that we can't observe cell division in cells of extinct species. This hypothesis uses only currently existing species as some possible examples of intermediate mechanisms. Other mechanisms may have existed in species that have gone extinct; we simply have no way of knowing.

Figure 12.13

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Mechanisms of cell division in several groups of organisms.
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Mechanisms of cell division shown for a. bacteria; b. dinoflagellates; c. diatoms and some yeasts and d. most eukaryotes.

*Some unicellular eukaryotes existing today have mechanisms of cell division that may resemble intermediate steps in the evolution of mitosis. Except for **(a)**, cell walls are not shown.*

Bacteria. During binary fission in bacteria, the origins of the daughter chromosomes move to opposite ends of the cell. The mechanism involves the polymerization of actin-like molecules, and possibly proteins that may anchor the daughter chromosomes to specific sites on the plasma membrane.

Dinoflagellates. In unicellular protists called dinoflagellates, the chromosomes attach to the nuclear envelope, which remains intact during cell division. Microtubules pass through the nucleus inside cytoplasmic tunnels, reinforcing the spatial orientation of the nucleus, which then divides in a process reminiscent of bacterial binary fission.

Diatoms and some yeasts. In these two other groups of unicellular eukaryotes, the nuclear envelope also remains intact during cell division. In these organisms, the microtubules form a spindle within the nucleus. Microtubules separate the chromosomes, and the nucleus splits into two daughter nuclei.

Most eukaryotes. In most other eukaryotes, including plants and animals, the spindle forms outside the nucleus, and the nuclear envelope breaks down during mitosis. Microtubules separate the chromosomes, and two nuclear envelopes then form.

Video

Nuclear Envelope Breakdown and Formation During Mitosis in *C. elegans*, a Eukaryote

Concept Check 12.2

- How many chromosomes are shown in the illustration in Figure 12.8? Are they duplicated? How many chromatids are shown?
- Compare cytokinesis in animal cells and plant cells.
- During which stages of the cell cycle does a chromosome consist of two identical chromatids?
- Compare the roles of tubulin and actin during eukaryotic cell division with the roles of tubulin-like and actin-like proteins during bacterial binary fission.
- A kinetochore has been compared to a coupling device that connects a motor to the cargo that it moves. Explain.
- **MAKE CONNECTIONS** What other functions do actin and tubulin carry out? Name the proteins they interact with to do so. (Review Figures 6.21a and 6.26a.)

*For suggested answers, see*Appendix A.**

Concept 12.3: The eukaryotic cell cycle is regulated by a molecular control system

The timing and rate of cell division in different parts of a plant or animal are crucial to normal growth, development, and maintenance. The frequency of cell division varies with the type of cell. For example, human skin cells divide frequently throughout life, whereas liver cells maintain the ability to divide but keep it in reserve until an appropriate need arises—say, to repair a wound. Some of the most specialized cells, such as fully formed nerve cells and muscle cells, do not divide at all in a mature human. These cell cycle differences result from regulation at the molecular level. The mechanisms of this regulation are of great interest, not only to understand the life cycles of normal cells but also to learn how cancer cells manage to escape the usual controls.

The Cell Cycle Control System

What controls the cell cycle? In the early 1970s, a variety of experiments led to the hypothesis that the cell cycle is driven by specific signaling molecules present in the cytoplasm. Some of the first strong evidence for this hypothesis came from experiments with mammalian cells grown in culture. In these experiments, two cells in different phases of the cell cycle were fused to form a single cell with two nuclei **(Figure 12.14)**. If one of the original cells was in the S phase and the other was in G1, the G1 nucleus immediately entered the S phase, as though stimulated by signaling molecules present in the cytoplasm of the first cell. Similarly, if a cell undergoing mitosis (M phase) was fused with another cell in any stage of its cell cycle, even G1, the second nucleus immediately entered mitosis, with condensation of the chromatin and formation of a mitotic spindle.

Figure 12.14

Inquiry: Do molecular signals in the cytoplasm regulate the cell cycle?

Experiment

Researchers at the University of Colorado wondered whether a cell's progression through the cell cycle is controlled by cytoplasmic molecules. They induced cultured mammalian cells at different phases of the cell cycle to fuse. Two experiments are shown.

Results

Two diagrams of experiments involving cells at different phases of the cell cycle.

Experiment 1: A cell drawn in S-Phase is fused to a cell in G 1 phase. The arrow shows these cells go to one large, fused cell with two nuclei both in S-phase and the D N A is synthesized.

Experiment 2: One cell drawn in M fused with another cell in G 1. The arrow shows the cells go to one large, fused cell with two nuclei both in M; the mitotic spindle is visible. However, the chromosome that had been in G 1 has entered mitosis even though it is unduplicated.

Conclusion

The results of fusing a G1 cell with a cell in the S or M phase of the cell cycle suggest that molecules present in the cytoplasm during the S or M phase control the progression to those phases.

Data from R. T. Johnson and P. N. Rao, Mammalian cell fusion: Induction of premature chromosome condensation in interphase nuclei, *Nature*226:717–722 (1970).

WHAT IF?*If the progression of phases did not depend on cytoplasmic molecules and each phase began when the previous one was complete, how would the results have differed?*

*For suggested answer, see***Appendix A*.

The experiment shown in Figure 12.14 and other experiments on animal cells and yeasts demonstrated that the sequential events of the cell cycle are directed by a distinct **cell cycle control system**, a cyclically operating set of molecules in the cell that both triggers and coordinates key events in the cell cycle **(Figure 12.15)**. The cell cycle control system has been compared to the control device of a washing machine. Like the washer's timing device, the cell cycle control system proceeds on its own, according to a built-in clock.

However, just as a washer's cycle is subject to both internal control (such as the sensor that detects when the tub is filled with water) and external adjustment (such as starting or stopping the machine), the cell cycle is regulated at certain checkpoints by both internal and external signals. A **checkpoint** in the cell cycle is a control point where stop and go-ahead signals can regulate the cycle. Three important checkpoints are found in the G1, G2, and M phases (red gates in Figure 12.15), which will be discussed shortly.

Figure 12.15

Mechanical analogy for the cell cycle control system.

Clock-like diagram of the cell cycle, with perimeter segregated into sections depicting sequential events.

In this diagram, the flat "stepping stones" around the perimeter represent sequential events. Like the control device of a washing machine, the cell cycle control system proceeds on its own, driven by a built-in clock. However, the system is subject to internal and external regulation at various checkpoints; three important checkpoints are shown (red).

Inner part of the clock is separated into pie-shaped slices representing G 1, S, G 2, and M phases. Arrows indicate a clockwise progression of these events. The cell's control system is separated by 3 checkpoints (red gates in the diagram): the G 1 checkpoint (located near the end of G 1), the G 2 checkpoint (located at the end of G 2), and the M checkpoint (located in the middle of the M slice).

Animation

Control of the Cell Cycle

image

Play

image

Play

To understand how cell cycle checkpoints work, we'll first identify some of the molecules that make up the cell cycle control system (the molecular basis for the cell cycle clock) and describe how a cell progresses through the cycle. We'll then consider the internal and external checkpoint signals that can make the clock either pause or continue.

The Cell Cycle Clock: Cyclins and Cyclin-Dependent Kinases

Rhythmic fluctuations in the abundance and activity of cell cycle control molecules pace the sequential events of the cell cycle. These regulatory molecules are mainly proteins of two types: protein kinases and

cyclins. As you learned in Concept 11.3, protein kinases are enzymes that activate or inactivate other proteins by phosphorylating them.

Many of the kinases that drive the cell cycle are actually present at a constant concentration in the growing cell, but much of the time they are in an inactive form. To be active, such a kinase must be attached to a **cyclin**, a protein that gets its name from its cyclically fluctuating concentration in the cell. Because of this requirement, these kinases are called **cyclin-dependent kinases**, or **Cdks**. The activity of a Cdk rises and falls with changes in the concentration of its cyclin partner. **Figure 12.16a** shows the fluctuating activity of **MPF**, the cyclin-Cdk complex that was discovered first (in frog eggs). Note that the peaks of MPF activity correspond to the peaks of cyclin concentration. The cyclin level rises during the S and G2 phases and then falls abruptly during M phase.

Figure 12.16

Molecular control of the cell cycle at the G2 checkpoint.

Molecular control of the cell cycle at the G 2 checkpoint.

Part a. Fluctuation of M P F activity and cyclin concentration during the cell cycle. The graph shows cyclin concentration and M P F activity build during S and G 2 phases and fall off sharply during the M phase. The graph shows 3 peaks for M P F activity and cyclin concentration (y-axis) over time (x-axis) during different phases of the cell cycle. During the first peak, cyclin concentration rises sharply before the M phase and then abruptly falls off during the M phase, while M P F activity has a symmetric (hump-shaped) increase and decrease around the same point. In the second peak, cyclin concentration rises sharply during the S and G 2 phases before falling abruptly during the M phase. M P F activity increases and decreases symmetrically before and during the M phase. The third peak follows the same pattern and peak 2.

Part b. Molecular mechanisms that help regulate the cell cycle.

- Synthesis of cyclin begins in the late S phase and continues through G 2. Because cyclin is protected from degradation during this stage, it accumulates.
- Cyclin combines with Cdk, producing M P F. When enough M P F molecules accumulate, the cell passes the G 2 checkpoint and begins mitosis.
- M P F promotes mitosis by phosphorylating various proteins. M P F‘s activity peaks during metaphase.
- During anaphase, the cyclin component of M P F is degraded, terminating the M phase. The cell enters the G 1 phase.
- 5. During G 1, the degradation of cyclin continues, and the Cdk component of M P F is recycled.

The steps of the cell cycle are timed by rhythmic fluctuations in the activity of cyclin-dependent kinases (Cdks). Here we focus on a cyclin-Cdk complex in animal cells called MPF, which acts at the G2 checkpoint as a go-ahead signal, triggering the events of mitosis.

VISUAL SKILLS *Explain how the events in the diagram in (b) are related to the "Time" axis of the graph in (a), beginning at the left.*

*For suggested answer, see*Appendix A.**

Interview

Interview with Paul Nurse: Discovering how protein kinases control the cell cycle

Paul Nurse.

The initials MPF stand for "maturation-promoting factor," but we can think of MPF as "M-phase-promoting factor" because it triggers the cell's passage into the M phase, past the G2 checkpoint. When cyclins that accumulate during G2 associate with Cdk molecules, the resulting MPF complex is active—it phosphorylates a variety of proteins, initiating mitosis **(Figure 12.16b)**. MPF acts both directly as a kinase and indirectly by activating other kinases. For example, MPF causes phosphorylation of various proteins of the nuclear lamina (see Figure 6.9), which promotes fragmentation of the nuclear envelope during prometaphase of mitosis. There is also evidence that MPF contributes to molecular events required for chromosome condensation and spindle formation during prophase.

During anaphase, MPF helps switch itself off by initiating a process that leads to the destruction of its own cyclin. The noncyclin part of MPF, the Cdk, persists in the cell, inactive until it becomes part of MPF again by associating with new cyclin molecules synthesized during the S and G2 phases of the next round of the cycle.

The fluctuating activities of different cyclin-Cdk complexes are of major importance in controlling all the stages of the cell cycle; they also give the go-ahead signals at some checkpoints. As mentioned above, MPF controls the cell's passage through the G2 checkpoint. Cell behavior at the G1 checkpoint is also regulated by the activity of cyclin-Cdk protein complexes. Animal cells appear to have at least three Cdk proteins and several different cyclins that operate at this checkpoint. Next, let's consider checkpoints in more detail.

Stop and Go Signs: Internal and External Signals at the Checkpoints

Animal cells generally have built-in stop signals that halt the cell cycle at checkpoints until overridden by go-ahead signals. Many signals registered at checkpoints come from cellular surveillance mechanisms inside the cell. These signals report whether crucial cellular processes that should have occurred by that point have in fact been completed correctly and thus whether or not the cell cycle should proceed. Checkpoints also register signals from outside the cell. The signals are transmitted within the cell by signal transduction

pathways (see Figure 11.6). Three important checkpoints are those in the G1, G2, and M phases (see Figure 12.15).

For many cells, the G1 checkpoint seems to be the most important. If a cell receives a go-ahead signal at the G1 checkpoint, it will usually complete the G1, S, G2, and M phases and divide. If it does not receive a go-ahead signal at that point, it may exit the cycle, switching into a nondividing state called the **G0 phase(Figure 12.17a)**. Most cells of the human body are actually in the G0 phase. As mentioned earlier, mature nerve cells and muscle cells never divide. Other cells, such as liver cells, can be "called back" from the G0 phase to the cell cycle by external cues, such as growth factors released during injury.

Figure 12.17

Two important checkpoints.

Two important checkpoints shown: Part a, G 1 checkpoint and part b, M checkpoint.

Part a. G 1 checkpoint. In the absence of a go-ahead signal, a cell exits the cell cycle and enters G 0, a non-dividing state.

- Diagram 1: Cell hits red gate at the G 1 checkpoint and does not pass, moving instead to the G0 phase. If a cell receives a go-ahead signal, the cell continues on in the cell cycle.
- Diagram 2: Cell passes the G 1 checkpoint and continues in the cell cycle.

Part b. M checkpoint. A cell in mitosis receives a stop signal when any of its chromosomes are not attached to spindle fibers.

- Diagram 1: A cell with chromosomes in prometaphase (most chromosomes are nearing the metaphase plate, but some remain distant) corresponds to the red gate at the M checkpoint in the cell cycle. The cell receives a stop signal. When all chromosomes are attached to spindle fibers from both poles, a go-ahead signal allows the cell to proceed into anaphase.
- Diagram 2: The cell with all chromosomes lined up at the metaphase plate corresponds with the M checkpoint in the cell cycle. The cell receives the go-ahead signal and proceeds to anaphase, with sister chromatids separated.

At certain checkpoints in the cell cycle (red gates), cells do different things depending on the signals they receive. Events of the (a) G1 and (**b**) M checkpoints are shown. In (**b**), the G2checkpoint has already been passed by the cell.

WHAT IF? *What might be the result if the cell ignored either checkpoint and progressed through the cell cycle?*

*For suggested answer, see**Appendix A.*

Biologists are currently working out the pathways that link signals originating inside and outside the cell with the responses by cyclin-dependent kinases and other proteins. An example of an internal signal occurs at the third important checkpoint, the M checkpoint **(Figure 12.17b)**. Anaphase, the separation of sister chromatids, does not begin until all the chromosomes are properly attached to the spindle at the metaphase plate. Researchers have learned that as long as some kinetochores are unattached to spindle microtubules, the sister chromatids remain together, delaying anaphase. Only when the kinetochores of all the chromosomes are properly attached to the spindle does the appropriate regulatory protein complex become activated. (In this case, the regulatory molecule is not a cyclin-Cdk complex but, instead, a different complex made up of several proteins.) Once activated, the complex sets off a chain of molecular events that activates the enzyme separase, which cleaves the cohesins, allowing the sister chromatids to separate. This mechanism ensures that daughter cells do not end up with missing or extra chromosomes.

There are checkpoints in addition to those in G1, G2, and M. For instance, a checkpoint in S phase stops cells with DNA damage from proceeding in the cell cycle. And, in 2014, researchers presented evidence for another checkpoint between anaphase and telophase that ensures anaphase is completed and the chromosomes are well separated before cytokinesis can begin, thus avoiding chromosomal damage.

What about the stop and go-ahead signals themselves—what are the signaling molecules? Studies using animal cells in culture have led to the identification of many external factors, both chemical and physical, that can influence cell division. For example, cells fail to divide if an essential nutrient is lacking in the culture medium. (This is analogous to trying to run a washing machine without the water supply hooked up; an internal sensor won't allow the machine to continue past the point where water is needed.) And even if all other conditions are favorable, most types of mammalian cells divide in culture only if the growth medium includes specific growth factors. As mentioned in Concept 11.1, a **growth factor** is a protein released by certain cells that stimulates other cells to divide. Different cell types respond specifically to different growth factors or combinations of growth factors.

Consider, for example, *platelet-derived growth factor* (*PDGF*), which is made by blood cell fragments called platelets. When an injury occurs, platelets release PDGF in the vicinity. The experiment illustrated in **Figure 12.18** demonstrates that PDGF is required for the division of cultured **fibroblasts**, a type of connective tissue cell. Fibroblasts have PDGF receptors on their plasma membranes. The binding of PDGF molecules to these receptors (which are receptor tyrosine kinases; see Figure 11.8) triggers a signal transduction pathway that allows the cells to pass the G1 checkpoint and divide. PDGF stimulates fibroblast division not only in the artificial conditions of cell culture but also in an animal's body. Thus, injury results in a proliferation of fibroblasts that helps heal the wound.

Figure 12.18

The effect of platelet-derived growth factor (PDGF) on cell division.

With platelet-derived growth factor (P D G F), cells proliferate. Without P D G F, cells fail to divide.

- A sample of human connective tissue is cut up into small pieces in a Petri dish using scalpels.
- Enzymes are used to digest the extracellular matrix in the tissue pieces, resulting in a suspension of free fibroblasts shown in a flask.
- Cells are transferred to culture vessels (rectangular bottle-like containers that can lie flat) containing a basic growth medium consisting of glucose, amino acids, salts, and antibiotics (to prevent bacterial growth).
- P D G F is added to half the vessels. The culture vessels are incubated at 37 degrees Celsius for 24 hours. Without P D G F (the control) in the basic growth medium, the cells fail to divide and few are shown in the culture vessel. With P D G F, in the basic growth medium, the cells proliferate. Many are shown in the culture vessel. An S E M shows cultured fibroblasts (scale: 10 micrometers).

MAKE CONNECTIONS *PDGF signals cells by binding to a cell-surface receptor tyrosine kinase. If you added a chemical that blocked phosphorylation, how would the results differ? (See*Figure 11.8*.)**

*For suggested answer, see*Appendix A*.**

The effect of an external physical factor on cell division is clearly seen in **density-dependent inhibition**, a phenomenon in which crowded cells stop dividing **(Figure 12.19a)**. Studies done many years ago showed that cultured cells normally divide until they form a single layer of cells on the inner surface of a culture flask, at which point the cells stop dividing. If some cells are removed, those bordering the open space begin dividing again and continue until the vacancy is filled. Follow-up studies revealed that the binding of a cell-surface protein to its counterpart on an adjoining cell sends a signal to both cells that inhibits cell division, preventing them from moving forward in the cell cycle, even in the presence of growth factors.

Figure 12.19

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    Density-dependent inhibition and anchorage dependence of
cell division.
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Density-dependent inhibition and anchorage dependence of cell division.

Individual cells are shown disproportionately large in the drawings.

Part a. Normal mammalian cells. Cell density is limited to a single layer by contact with neighboring cells and the availability of nutrients, growth factors, and a substratum for attachment.

First culture vessel: Cells anchor to the dish surface and divide (anchorage dependence). Few cells are shown.

Second culture vessel: When cells have formed a complete single layer, they stop dividing (density-dependent inhibition). Cells pictured in the culture vessel drawing take up only a single layer.

Third culture vessel: If some cells are scraped away, the remaining cells divide to fill the gap and then stop once they contact each other (density-dependent inhibition). The drawing shows a small patch of cells being scraped away inside the culture vessel.

Fourth culture vessel: A single layer of cells again fills the vessel. The image zooms in on cells. Scale: 20 micrometers.

Part b. Cancer cells. Cancer cells usually continue to divide well beyond a single layer, forming a clump of overlapping cells. They do not exhibit anchorage dependence or density-dependent inhibition.

Most animal cells also exhibit **anchorage dependence** (see Figure 12.19a). To divide, they must be attached to something, such as the inside of a culture flask or the extracellular matrix of a tissue. Experiments suggest that, like cell density, anchorage is signaled to the cell cycle control system via pathways involving plasma membrane proteins and elements of the cytoskeleton linked to them.

Density-dependent inhibition and anchorage dependence appear to function not only in cell culture but also in the body's tissues, checking the growth of cells at some optimal density and location during embryonic development and throughout an organism's life. Cancer cells, which we examine next, exhibit neither density-dependent inhibition nor anchorage dependence **(Figure 12.19b)**.

Loss of Cell Cycle Controls in Cancer Cells

Cancer cells do not heed the normal signals that regulate the cell cycle. In culture, they do not stop dividing when growth factors are depleted. A logical hypothesis is that cancer cells do not need growth factors in their culture medium to grow and divide. They may make a required growth factor themselves, or they may have an abnormality in the signaling pathway that conveys the growth factor's signal to the cell cycle control system even in the absence of that factor. Another possibility is an abnormal cell cycle control system. In these scenarios, the underlying basis of the abnormality is almost always a change in one or more genes (for example, a mutation) that alters the function of their protein products, resulting in faulty cell cycle control. We'll explore the molecular basis for such changes in Concept 18.5.

There are other important differences between normal cells and cancer cells that reflect malfunctions of the cell cycle. If and when they stop dividing, cancer cells do so at random points in the cycle, rather than at the normal checkpoints. Moreover, cancer cells can go on dividing indefinitely in culture if they are given a continual supply of nutrients; in essence, they are "immortal." A striking example is a cell line that has been reproducing in culture since 1951. Cells of this line are called HeLa cells because their original source was a tumor removed from a woman named Henrietta Lacks. (Neither Ms. Lacks nor her family gave permission

or even knew about the propagation and use of her cells, which have helped biologists make countless significant discoveries over the years.) Cells in culture that acquire the ability to divide indefinitely are said to have undergone a process called **transformation**, causing them to behave (in cell division, at least) like cancer cells. By contrast, nearly all normal, nontransformed mammalian cells growing in culture divide only about 20 to 50 times before they stop dividing, age, and die. Finally, cancer cells evade the normal controls that trigger a cell to undergo apoptosis when something is wrong—for example, when an irreparable mistake has occurred during DNA replication preceding mitosis.

Abnormal cell behavior in the body can be catastrophic. The problem begins when a single cell in a tissue undergoes the first of many steps that convert a normal cell to a cancer cell. Such a cell often has altered proteins on its surface, and the body's immune system normally recognizes the cell as "nonself"—an insurgent—and destroys it. However, if the cell evades destruction, it may proliferate and form a tumor, a mass of abnormal cells within otherwise normal tissue. The abnormal cells may remain at the original site if their genetic and cellular changes don't allow them to move to or survive at another site. In that case, the tumor is called a **benign tumor**. Most benign tumors do not cause serious problems (depending on their location) and can be removed by surgery. In contrast, a **malignant tumor** includes cells whose genetic and cellular changes enable them to spread to new tissues and impair the functions of one or more organs; these cells are also sometimes called *transformed* cells (although usage of this term is generally restricted to cells in culture). An individual with a malignant tumor is said to have cancer **(Figure 12.20)**.

Figure 12.20

The growth and metastasis of a malignant breast tumor.

The growth and metastasis of a malignant breast tumor.

A series of genetic and cellular changes contribute to a tumor becoming malignant (cancerous). The cells of malignant tumors grow in an uncontrolled way and can spread to neighboring tissues and, via lymph and blood vessels, to other parts of the body (metastasis).

- A tumor grows from a single cancer cell. The tumor is pictured as a mass of cells in a drawing of a human breast with glandular tissue labeled.
- Cancer cells invade neighboring tissue. The mass of cells has grown and now invades glandular tissue.
- Cancer cells spread through lymph and blood vessels to other parts of the body. Tumor cells leave the breast tissue and travel in both lymph and blood vessels to other parts of the body.
- A small percentage of cancer cells may metastasize to another part of the body. The drawing shows a metastatic tumor growing in a distant part of the body as a result of cancer cells traveling via the blood and lymph vessels. Breast cancer cells are shown in a colorized S E M (scale: 5 micrometers).

HHMI BioInteractive Animation

How Tumors Grow

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

The changes that have occurred in cells of malignant tumors show up in many ways besides excessive proliferation. These cells may have unusual numbers of chromosomes, though whether this is a cause or an effect of tumor-related changes is an ongoing debate. Their metabolism may be altered, and they may cease to function in any constructive way. Abnormal changes on the cell surface cause cancer cells to lose attachments to neighboring cells and the extracellular matrix, allowing them to spread into nearby tissues. Cancer cells may also secrete signaling molecules that cause blood vessels to grow toward the tumor. A few tumor cells may separate from the original tumor, enter blood vessels and lymph vessels, and travel to other parts of the body. There, they may proliferate and form a new tumor. This spread of cancer cells to locations distant from their original site is called **metastasis** (see Figure 12.20).

A tumor that appears to be localized may be treated with high-energy radiation, which damages DNA in cancer cells much more than DNA in normal cells, apparently because the majority of cancer cells have lost the ability to repair such damage. To treat known or suspected metastatic tumors, chemotherapy is used, in which drugs that are toxic to actively dividing cells are administered through the circulatory system. As you might expect, chemotherapeutic drugs interfere with specific steps in the cell cycle. For example, the drug Taxol freezes the mitotic spindle by preventing microtubule depolymerization, which stops actively dividing cells from proceeding past metaphase and leads to their destruction. The side effects of chemotherapy are due to the effects of the drugs on normal cells that divide frequently, due to the function of that cell type in the organism. For example, nausea results from chemotherapy's effects on intestinal cells, hair loss from effects on hair follicle cells, and susceptibility to infection from effects on immune system cells. You can work with data from an experiment involving a potential chemotherapeutic agent in the **Scientific Skills Exercise**.

Scientific Skills Exercise

Interpreting Histograms

AP Science Practices 1, 2, 4, 6

At What Phase Is the Cell Cycle Arrested by an Inhibitor? Many medical treatments are aimed at stopping cancer cell proliferation by blocking the cell cycle of cancerous tumor cells. One potential treatment is a cell cycle inhibitor derived from human umbilical cord stem cells. In this exercise, you will compare two histograms to determine where in the cell cycle the inhibitor blocks the division of cancer cells.

Human glioblastoma glows green.

Human brain cancer cell

How the Experiment Was Done In the treated sample, human glioblastoma (brain cancer) cells were grown in tissue culture in the presence of the inhibitor, while a control sample of glioblastoma cells was grown in its absence. After 72 hours of growth, the two cell samples were harvested. To get a "snapshot" of the phase of the cell cycle each cell was in at that time, the samples were treated with a fluorescent chemical that binds to DNA and then run through a flow cytometer, an instrument that records the fluorescence level of each cell. Computer software then graphed the number of cells in each sample with a particular fluorescence level, as shown below.

Data from the Experiment

Histograms showing the y-axis has number of cells from 0 to 200 in increments of 40 units and the x-axis has amount of fluorescence per cell (fluorescence units) from 0 to 600 in increments of 200 units, for the Control and the Treated samples.

In each case the histograms are broken into 3 sections. For the Control samples, section A starts at 150, 10, peaks at 200, 130 and ends at 250, 20; section B starts at 250, 20, drops to 275, 15, and ends at 325, 20; section C starts at 325, 20, peaks at 375, 40 and drops to 425, 5. For the Treated samples, section A starts at 150, 0, peaks at 200, 170, and ends at 250, 20; section B starts at 250, 20, drops to 275, 10, and ends at 325, 20; section C starts at 325, 20 and drops to 425, 5.

Data from K. K. Velpula et al., Regulation of glioblastoma progression by cord blood stem cells is mediated by downregulation of cyclin D1, *PLoS ONE* 6(3):e18017 (2011).

The data are plotted in a type of graph called a histogram (above), which groups the values for a numeric variable on the x-axis into intervals. A histogram allows you to see how all the experimental subjects (cells, in this case) are distributed along a continuous variable (amount of fluorescence). In these histograms, the bars are so narrow that the data appear to follow a curve for which you can detect peaks and dips. Each narrow bar represents the number of cells observed to have a level of fluorescence in the range of that interval. This in turn indicates the relative amount of DNA in those cells. Overall, comparing the two histograms allows you to see how the DNA content of this cell population is altered by the treatment.

INTERPRET THE DATA

- Study the data in the histograms.
- Which axis indirectly shows the relative amount of DNA per cell? Explain your answer.
- In the control sample, compare the first peak in the histogram (in region A) to the second peak (in region C). Which peak shows the population of cells with the higher amount of DNA per cell? Explain. (For additional information about histograms, see the Scientific Skills Review in Appendix D.)
-
    - In the control sample histogram, identify the phase of the cell cycle (G1, S, or G2) of the population of cells in each region delineated by vertical lines. Label the histogram with these phases and explain your answer.
- Does the S phase population of cells show a distinct peak in the histogram? Why or why not?
- The histogram representing the treated sample shows the effect of growing the cancer cells alongside human umbilical cord stem cells that produce the potential inhibitor.
- Label the histogram with the cell cycle phases. Which phase of the cell cycle has the greatest number of cells in the treated sample? Explain.
- Compare the distribution of cells among G1, S, and G2 phases in the control and treated samples. What does this tell you about the cells in the treated sample?
- Based on what you learned in Concept 12.3, propose a mechanism by which the stem cell–derived inhibitor might arrest the cancer cell cycle at this stage. (More than one answer is possible.)

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Over the past several decades, researchers have produced a flood of valuable information about cell-signaling pathways and how their malfunction contributes to the development of cancer through effects on the cell cycle. Coupled with new molecular techniques, such as the ability to rapidly sequence the DNA of cells in a particular tumor, medical treatments for cancer are beginning to become more “personalized” to a particular patient’s tumor.

BBC Video

Are Fruit Flies the Key in the Fight Against Cancer?

image

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image

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For example, the cells of roughly 20% of breast cancer tumors show abnormally high amounts of a cell-surface receptor tyrosine kinase called HER2, and many show an increase in the number of estrogen receptor (ER) molecules, intracellular receptors that can trigger cell division. Based on lab findings, a physician can prescribe chemotherapy with a molecule that blocks the function of the specific protein (Herceptin for HER2 and tamoxifen for ERs). Treatment using these agents, when appropriate, has led to increased survival rates and fewer cancer recurrences (see Make Connections Figure 18.27).

Interview

Interview with Bruce Alberts: Cancer control and careers in science

A photo of Bruce Alberts.

Concept Check 12.3

- In Figure 12.14, why do the nuclei resulting from experiment 2 contain different amounts of DNA?
- How does MPF allow a cell to pass the G2 phase checkpoint and enter mitosis? (See Figure 12.16.)
- **MAKE CONNECTIONS** Explain how receptor tyrosine kinases and intracellular receptors might function in triggering cell division. (Review Figures 11.8 and 11.9 and Concept 11.2.)

*For suggested answers, see*Appendix A.**